NOETHERIANITY, SURJECTIVITY AND FINITE RESIDUE EXTENSIONS FOR THE INTEGRAL CLOSURE IN A BASE FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 35925-35935 (canonical tag 09IH) and the counted proof is the proof environment at source lines 35937-35958, which immediately follows it and is its own proof.

Editorial note. The printed text writes suboject in the section introduction rather than in this proof; inside the counted spans the wording is copied exactly as printed and no correction is applied.

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Allowed standard prerequisites (editorial).

- A localization of a finite type algebra at a prime is integral over the localization of the finite type algebra after an integral base change, so dimensions agree. used at proof lines 35937-35945. Shows that the localized finite type algebra over the new base is one dimensional and Noetherian, which is the setting for the integral closure.
- Krull-Akizuki: the integral closure of a one dimensional Noetherian domain in a finite extension of its fraction field is Noetherian. invoked at proof lines 35948-35952. Makes each of the finitely many integral closures of the quotients Noetherian.
- For a one dimensional local ring the residue field extension in a finite extension of the fraction field is finite. invoked at proof lines 35952-35955. Gives the finiteness of the residue field extensions asserted in the statement.

Source reference. Original lemma 00OK: sources/stacks-chow/tags.txt, source lines 898–898. The reference is applied at proof lines 35944-35945 to conclude that the localized ring has dimension one, dimensions being preserved under integral extensions; the list of minimal primes that follows is inside the counted span.

Source reference. Original lemma 00PG: sources/stacks-chow/tags.txt, source lines 930–930. The reference gives Noetherianity of the integral closure of a one dimensional Noetherian domain in a finite extension, applied at proof lines 35948-35952

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to each quotient; the identification of the product with the integral closure is inside the counted span.

Source reference. Original lemma 031F: sources/stacks-chow/tags.txt, source lines 3982–3982. The reference supplies the finiteness of the residue field extensions for one dimensional rings, applied at proof lines 35952-35955; it is the last input and the conclusion drawn from it is inside the counted span.

Lemma 0.2. *Let $A \to B$ be an extension of discrete valuation rings with fraction fields $K \subset L$. Assume B is essentially of finite type over A. Let K'/K be an algebraic extension of fields such that the integral closure A' of A in K' is Noetherian. Then the integral closure B' of B in $L' = (L \otimes_K K')_{red}$ is Noetherian as well. Moreover, the map* $\mathrm{Spec}(B') \to \mathrm{Spec}(A')$ *is surjective and the corresponding residue field extensions are finitely generated field extensions.*

Proof. Let $A \to C$ be a finite type ring map such that B is a localization of C at a prime $\mathfrak{p}$. Then $C' = C \otimes_A A'$ is a finite type A'-algebra, in particular Noetherian. Since $A \to A'$ is integral, so is $C \to C'$. Thus $B = C_{\mathfrak{p}} \subset C'_{\mathfrak{p}}$ is integral too. It follows that the dimension of $C'_{\mathfrak{p}}$ is 1 (Algebra, Lemma 00OK). Of course $C'_{\mathfrak{p}}$ is Noetherian. Let $\mathfrak{q}_1, \ldots, \mathfrak{q}_n$ be the minimal primes of $C'_{\mathfrak{p}}$. Let B'_i be the integral closure of $B = C_{\mathfrak{p}}$, or equivalently by the above of $C'_{\mathfrak{p}}$ in the field of fractions of $C'_{\mathfrak{p}'}/\mathfrak{q}_i$. It follows from Krull-Akizuki (Algebra, Lemma 00PG applied to the finitely many localizations of $C'_{\mathfrak{p}}$ at its maximal ideals) that each B'_i is Noetherian. Moreover the residue field extensions in $C'_{\mathfrak{p}} \to B'_i$ are finite by Algebra, Lemma 031F. Finally, we observe that $B' = \prod B'_i$ is the integral closure of B in $L' = (L \otimes_K K')_{red}$. □